MILITARY SPECIFICATION

MECHANISM, RECOIL, 155MM HOWITZER: M45

This amendment forms a part of Military Specification MIL-M-45986 (WC), dated 15 February 1977, and is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 6

* 3.3.2 Function firing: Delete in its entirety and substitute:

"3.3.2 Function firing. The recoil mechanism shall withstand the function firing test (powder gymnastication) as specified in 4.6.1 without evidence of malfunction or failure and shall meet the criteria of 3.3.2.1 through 3.3.2.5. After function firing, the recoil mechanism shall meet requirements specified in 3.3.3 through 3.3.7.

3.3.2.1 Long recoil length. Maximum long recoil length shall not exceed 71 inches.

3.3.2.2 Short recoil length. Maximum short recoil length shall not exceed 50 inches.

3.3.2.3 Cycle time. Recoil cycle time shall not be less than 2 seconds and not more than 4 seconds.

3.3.2.4 Terminal velocity. At a distance of 2 inches from battery position, the counter recoil terminal velocity shall not exceed 4 inches/seconds.

3.3.2.5 Rod pull curves. The actual rod curves shall meet the following requirements: (see Figure 1)

Range 1: Peak rod pull shall occur within the first 20 milliseconds for long recoil and 60 milliseconds for short recoil.

Range 2: Within 280 miliseconds for long recoil or 180 miliseconds for short recoil, the continuous rod pull curve shall not exceed the peak rod pull of range 1.

Range 3: After Range 2, the rod pull curve shall be continuously declining to the end of the recoil cycle.

Range 4: Counter recoil cycle. See paragraph 3.3.2.4."

FSC 1025

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:34Z
Check the source to verify that this is the current version before use.

* 4.6.1 Function firing test: Delete in its entirety and substitute:

"4.6.1 Function firing test. Each recoil mechanism shall be subjected to the powder gymnastication test as specified in Table IV. All testing shall be done on a test stand at zero degrees elevation with the variable recoil locked for long recoil or short recoil, as specified. The tabulated sequence must be followed. Charges shall be varied as required to meet the stated rod pull requirements and shall impart an impulse of at least 10,000 pounds-seconds to at least one long recoil test mode and the short recoil test mode. If any of the requirements of paragraphs 3.3.2.1 through 3.3.2.5 are not met, the recoil mechanism shall be rejected. Upon completion of this test, the recoil mechanism shall be subjected to the tests specified in paragraphs 4.6.2 through 4.6.7.

Table IV Function Firing

Sequence #	Variable Recoil Setting	Total Required Rod Pull (Lbs)
1	Long Recoil	Warmer
2	Long Recoil	Max. Rod Pull
3	Short Recoil	(Total of both
4	Long Recoil	recoil rods) shall not exceed 82,000 pound-force.

PAGE 16

* Delete and add new paragraph as follows:

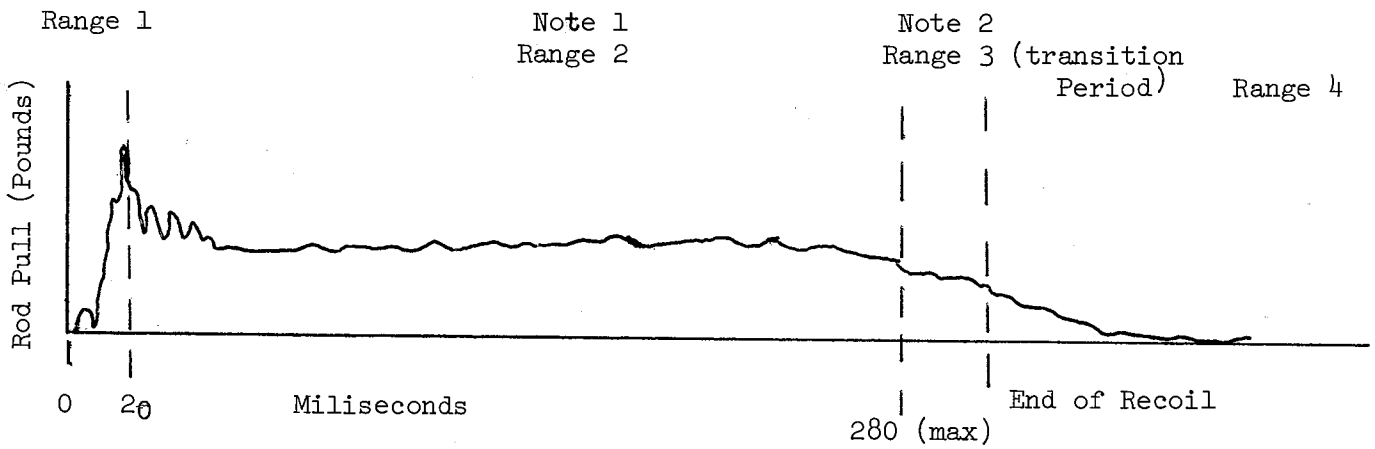
"6.3 One (1) copy of the firing report, which shall include variable data received by this specification and a digitized computer plot of rod pull versus time for at least the first 300 milliseconds of the recoil cycle, shall be sent to DRDAR-QAR-Q and DRDAR-LCW-A, ARRADCOM, Dover, NJ 07801 for informational purposes."

Delete original Figure 1 and substitute new Figure 1.

NOTE: The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:34Z
Check the source to verify that this is the current version before use.

LONG RECOIL



Note 1: Rod pull trace is allowed to increase or decrease within the range as long as it does not exceed the peak in Range 1.

Note 2: Rod pull trace must be continuously decreasing in this range to the end of the recoil cycle.

SHORT RECOIL

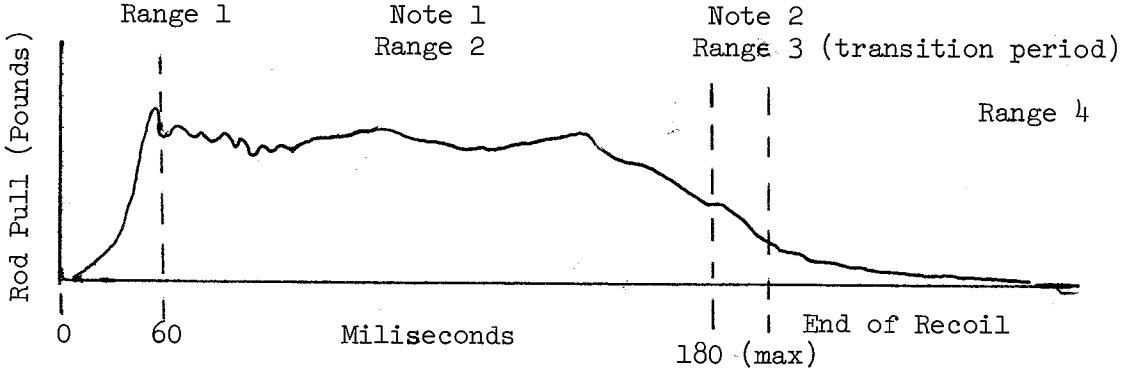


FIGURE 1 TYPICAL ROD PULL CURVES

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:34Z
Check the source to verify that this is the current version before use.

Custodian:
Army-AR

Preparing Activity:
Army-AR

(Project 1025-A132

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:34Z
Check the source to verify that this is the current version before use.